

Mathematics for New Technologies in Finance: Project

Hugo Belzil – Spring 2026, ETH Zürich

PART I.

Q1. The set Θ_0 consists of strategies $\theta_t = \nabla\varphi(X_t)$, where X_t is the traded asset and $\varphi \in \mathcal{D}$. Thus $\varphi : E \subset \mathbb{R}^d \rightarrow \mathbb{R}$ is twice continuously differentiable on E and satisfies

$$\int_E \left| \frac{\text{Tr}(A(x) \nabla^2 e^{\varphi(x)})}{e^{\varphi(x)}} \right| dx < \infty, \quad A(x) = c p(x)$$

where $c \in \mathbb{S}^d$ is symmetric positive definite and p is the density of $\mathcal{N}(0, \Sigma)$.

Q2. We start from the the stochastic exponential and get,

$$\log V_T^{\theta_0} = \int_0^T \nabla\varphi(X_t) dX_t - \frac{1}{2} \left[\int_0^T \nabla\varphi(X_t) dX_t \right]_T$$

where continuity of $\nabla\varphi$ and X_t trajectories lets us identify optional and quadratic variation. Itô's formula gives

$$d\varphi(X_t) = \nabla\varphi(X_t) dX_t + \frac{1}{2} \text{Tr}(\nabla^2\varphi(X_t) d[X]_t), \quad d[X^i, X^j]_t = c_{ij} dt$$

Plugging this into the previous display yields

$$\log V_T^{\theta_0} = \varphi(X_T) - \varphi(X_0) - \frac{1}{2} \int_0^T \text{Tr}\{c(\nabla^2\varphi(X_t) + \nabla\varphi(X_t) \nabla\varphi(X_t)^\top)\} dt$$

Since

$$\frac{\nabla^2 e^{\varphi(x)}}{e^{\varphi(x)}} = \nabla^2\varphi(x) + \nabla\varphi(x) \nabla\varphi(x)^\top$$

and since the OU process admits invariant law $\mathcal{N}(0, \Sigma)$ with density p , the ergodic property says that the long-run time average can be replaced by the expectation under p . Therefore

$$\lim_{T \rightarrow \infty} \frac{1}{T} \log V_T^{\theta_0} = -\frac{1}{2} \int_E \frac{\text{Tr}(c \nabla^2 e^{\varphi(x)})}{e^{\varphi(x)}} p(x) dx = -\frac{1}{2} \int_E \frac{\text{Tr}(A(x) \nabla^2 e^{\varphi(x)})}{e^{\varphi(x)}} dx$$

Q3. Let $u = e^\varphi$. For fixed (i, j) , integration by parts gives a boundary term $\int_{\partial E} A_{ij} \partial_i u u^{-1} n_j dS$, which vanishes by compact support/smoothness, and an interior term $-\int_E \partial_j (A_{ij}/u) \partial_i u dx$. Hence

$$\begin{aligned} \int_E \frac{\text{Tr}(A \nabla^2 u)}{u} dx &= \sum_{i,j} \int_E \frac{A_{ij}}{u} \partial_{ij} u dx = \sum_{i,j} \int_E [-\partial_j A_{ij} \partial_i \varphi + A_{ij} \partial_i \varphi \partial_j \varphi] dx \\ &= -\int_E (\text{Div } A)^\top \nabla\varphi dx + \int_E \nabla\varphi^\top A \nabla\varphi dx \end{aligned}$$

In the last line we only rewrote the double sums in vector notation, using the cyclic invariance of the trace.

Q4. For a matrix field, $(\text{Div } A)_i = \sum_{j=1}^d \partial_j A_{ij}$. Since $A = cp$ and c is constant,

$$(\text{Div } A)_i = \sum_j c_{ij} \partial_j p = [c]_i \cdot \nabla p, \quad \text{Div } A = c \nabla p, \quad A^{-1} \text{Div } A = \frac{1}{p} \nabla p = \nabla \log p$$

Q5. From Q4, $\text{Div } A = A \nabla \log p$, and for p Gaussian, $\nabla \log p = -\Sigma^{-1}x$. Therefore Q3 becomes

$$\int_E \frac{\text{Tr}(A \nabla^2 e^\varphi)}{e^\varphi} dx = \int_E (x^\top \Sigma^{-1} + \nabla\varphi^\top) A \nabla\varphi dx$$

Since on Θ_0 the asymptotic growth rate is reduced to the variational expression above, after rewriting the sup as an inf problem, the problem becomes

$$\begin{aligned} \lambda^* &= \inf_{\varphi \in \mathcal{D}} \frac{1}{2} \int_E (x^\top \Sigma^{-1} + \nabla\varphi^\top) A \nabla\varphi dx \\ &= \inf_{\varphi \in \mathcal{D}} \frac{1}{2} \int_E \left[\left(\nabla\varphi + \frac{1}{2} \Sigma^{-1} x \right)^\top A \left(\nabla\varphi + \frac{1}{2} \Sigma^{-1} x \right) - \frac{1}{4} (\Sigma^{-1} x)^\top A (\Sigma^{-1} x) \right] dx \end{aligned}$$

Since A is positive definite and the last term is independent of φ , the minimizer satisfies $\nabla\varphi^*(x) = -\frac{1}{2} \Sigma^{-1} x$.

Q6. We conclude from question 5 that the robust-optimal strategy is

$$\theta_t^* = \nabla \varphi^*(X_t) = -\frac{1}{2} \Sigma^{-1} X_t$$

Q7. Evaluating the completed-square expression at θ^* gives

$$\lambda^* = \frac{1}{8} \int_E (\Sigma^{-1} x)^\top A(x) (\Sigma^{-1} x) dx = \frac{1}{8} \mathbb{E}[X^\top \Sigma^{-1} c \Sigma^{-1} X]$$

For $X_\infty \stackrel{d}{=} X \sim \mathcal{N}(0, \Sigma)$, $\mathbb{E}[XX^\top] = \Sigma$, so the trace identity gives

$$\lambda^* = \frac{1}{8} \text{Tr}(\Sigma^{-1} c \Sigma^{-1} \Sigma) = \frac{1}{8} \text{Tr}(c \Sigma^{-1})$$

PART II.

Q1. The d -dimensional OU dynamics are $dX_t = B(m - X_t)dt + c^{1/2}dW_t$, with $c^{1/2}(c^{1/2})^\top = c$. Since Part I requires invariant law $N(0, \Sigma)$, the long-run mean is $m = 0$, hence

$$dX_t = -BX_t dt + c^{1/2}dW_t$$

In the minimax setup, Nature only acts through the drift matrix B , since the diffusion matrix c is fixed.

Q2. For $dX_t = -BX_t dt + \sigma dW_t$, $c = \sigma \sigma^\top$, the density $P(x, t)$ of the random variable X_t must satisfy the Fokker-Planck equation, given by:

$$\partial_t P(x, t) = \sum_{i,j} B_{ij} \partial_{x_i} (x_j P) + \frac{1}{2} \sum_{i,j} (\sigma \sigma^\top)_{ij} \partial_{x_i x_j} P$$

At stationarity, $P = p$ is the Gaussian density of $N(0, \Sigma)$ which implies that $\partial_t P = 0$, so rewriting the FP PDE gives

$$\text{Tr}(B)p + (\nabla p)^\top Bx + \frac{1}{2} \text{Tr}(c \nabla^2 p) = 0$$

Using $\nabla p = -\Sigma^{-1} x p$ and $\nabla^2 p = (\Sigma^{-1} x x^\top \Sigma^{-1} - \Sigma^{-1})p$, this is equivalent to the Lyapunov condition

$$B\Sigma + \Sigma B^\top = c$$

(To avoid long computations to appear here, we explain briefly: the Lyapunov condition arises after using the fact that any real matrix can be decomposed into the sum of a symmetric matrix and a skew-symmetric one). Thus Nature must choose B subject to this constraint in order to preserve the invariant distribution.

Q3. The symmetry of c and the symmetric/skew-symmetric decomposition of ΣB imply

$$B = \left(\frac{1}{2} c + A \right) \Sigma^{-1}$$

where A is any $d \times d$ skew-symmetric matrix, i.e. $A^\top = -A$. Such a matrix has zero diagonal and opposite off-diagonal entries, so only the strictly upper-triangular entries are free: $d(d-1)/2$ degrees of freedom. For $d = 2$ we therefore use

$$B_\beta = \left(\frac{1}{2} c + \beta_{\text{skew}} \right) \Sigma^{-1} \quad \beta_{\text{skew}} = \begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix}$$

Q4. The Agent is an MLP $\varphi_\alpha : \mathbb{R}^2 \rightarrow \mathbb{R}$ with architecture $2 \rightarrow 64 \rightarrow 64 \rightarrow 64 \rightarrow 1$ and tanh activations; its strategy is $\theta_\alpha(x) = \nabla_x \varphi_\alpha(x)$. Since $d = 2$, Nature is represented by the single bounded (see implementation) trainable scalar β in Q3, implemented through a tanh transform, which automatically enforces $B_\beta \Sigma + \Sigma B_\beta^\top = c$. For samples $x^{(i)} \sim N(0, \Sigma)$, we estimate

$$\hat{g}(\alpha, \beta) = \frac{1}{M} \sum_{i=1}^M \left[-\theta_\alpha(x^{(i)})^\top B_\beta x^{(i)} - \frac{1}{2} \theta_\alpha(x^{(i)})^\top c \theta_\alpha(x^{(i)}) \right]$$

This uses the ergodic formulation from Part I rather than simulating full paths and discretizing stochastic integrals. The Agent maximizes $\hat{g}$, Nature minimizes it, so $\mathcal{L}_{\text{Agent}} = -\hat{g}$ and $\mathcal{L}_{\text{Nature}} = \hat{g}$, with alternating gradient steps.

Q5. We used the losses from Q4. To stabilize training, we used 5000 epochs, a large Monte Carlo sample of size $M = 8192$ to reduce gradient variance at each gradient step, and small learning rates ($\eta_{\text{nature}} = 5 \cdot 10^{-4}$ and $\eta_{\text{agent}} = 10^{-3}$).

Q6. The first plot shows convergence of the approximated growth rate to λ^* after a few hundred epochs. The second and third plots compare the Agent vector field with the closed-form strategy $-0.5\Sigma^{-1}X_t$. They are very similar with the largest errors near the bottom-left and top-right corners of $[-3, 3]^2$.

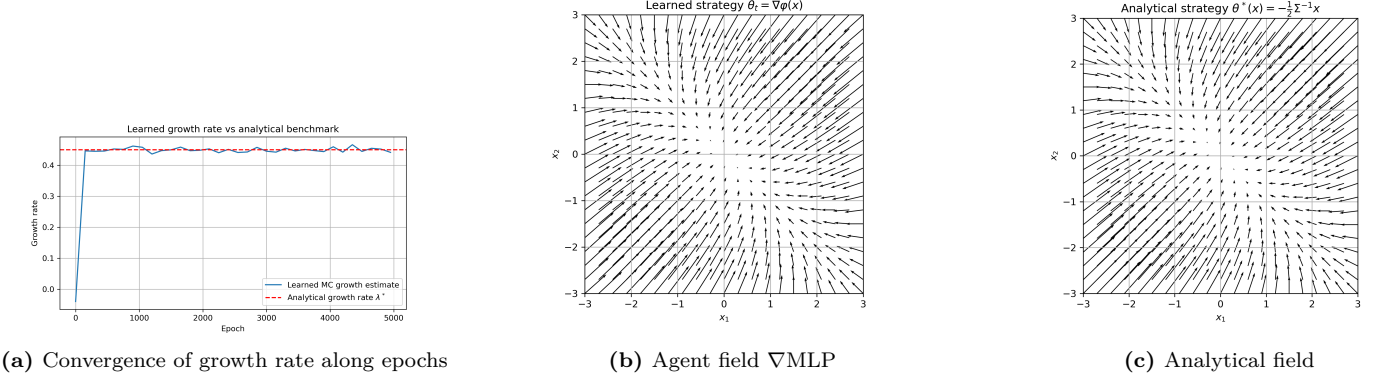


Figure 1: Numerical results from training.

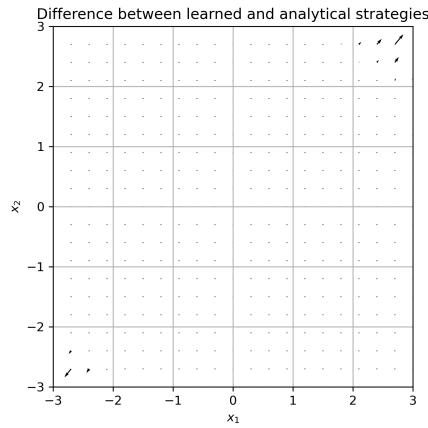


Figure 2: Vector field $\nabla\text{MLP} + \frac{1}{2}\Sigma^{-1}x$

Figure 2 shows the difference of the approximated versus analytical fields, which is almost constantly zero on the square $[-3, 3]^2$. With our choice of Σ , we have $\mathbb{P}(X_\infty \notin [-3, 3]^2) \approx 0.52\%$

Q7. On a 20K-point sample from N_∞ , we obtain $\widehat{\lambda^*} = 0.448799$, versus the closed form $\lambda^* = 0.45$ from the chosen c and Σ . This gives a relative error of about 0.2%, which is reasonable for the Monte Carlo/training approximation, with training taking about 1min30s on a local PC. In conclusion, the adversarial experiment has been able to recover accurately both the optimal strategy as well as the asymptotic growth rate derived analytically in part I.

Final notes

The code (self-contained Jupyter notebook) as well as the reference paper used for this mini-project are available on [GitHub](#). Random seeds have been fixed during experiments to ensure reproducibility. This repository also contains a slightly longer version of this project containing more details about the computations.